A Project Report on

Automatic Software Development with AI

Submitted in partial fulfillment for award of

Bachelor of Technology

Degree

in

CSE (Artificial Intelligence and Machine Learning)

By

A. Sravani (Y22ACM406) **A. Srilakshmi (Y22ACM402)**

A. Sai (Y22ACM405) **K. Venkata Krishna (Y22ACM447)**

T. Nagendra (L23ACM501)

Under the guidance of

Mr. N. Venkateswara Rao, **M. Tech**

Assistant Professor, Department of CSE (AIML)

Department of CSE **(Artificial Intelligence and Machine Learning)**

Bapatla Engineering College

(Autonomous)

(Affiliated to Acharya Nagarjuna University)

BAPATLA – 522 102, Andhra Pradesh, INDIA 2025-2026

Department of

CSE (Artificial Intelligence and Machine Learning)

CERTIFICATE

This is to certify that the project report entitled **Automatic Software Development with AI** that is being submitted by A. Sravani (Y22ACM406), A. Srilakshmi (Y22ACM402), A. Sai (Y22ACM405), K. Venkata Krishna (Y22ACM447), T. Nagendra (L23ACM501) in partial fulfillment for the award of the Degree of Bachelor of Technology in Computer Science & Engineering (Artificial Intelligence and Machine Learning) to the Acharya Nagarjuna University is a record of bonafide work carried out by them under our guidance and supervision.

Date:

Signature of the Guide
Mr. N. Venkateswara Rao, M.Tech,
Assistant Professor

Signature of the HOD
Dr. K. Mani Deep
Assoc. Prof. & Head

DECLARATION

We declare that this project work is composed by ourselves, that the work contained herein is our own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

[A. Sravani (Y22ACM406)]

[A. Srilakshmi (Y22ACM402)]

[A. Sai (Y22ACM405)]

[K. Venkata Krishna (Y22ACM447)]

[T. Nagendra (L23ACM501)]

Acknowledgement

We sincerely thank the following distinguished personalities who have given their advice and support for successful completion of the work.

We are deeply indebted to our most respected guide **Mr. N. Venkateswara Rao**, Assistant Professor, Department of CSE (AIML), Bapatla Engineering College, for his valuable and inspiring guidance, comments, suggestions and encouragement throughout the development of this project.

We extend our sincere thanks to **Dr. K. Mani Deep**, Head of the Department, for extending his cooperation and providing the required resources.

We would like to thank our beloved Principal **Dr. B. Srinivasa Rao** for providing the online resources and other facilities to carry out this work.

We would like to sincerely thank our project coordinator **Ms. G.V. Leela Kumari**, Assistant Professor, Dept. of CSE (AIML), for her helpful suggestions in presenting this document

We extend our sincere thanks to all other teaching faculty and non-teaching staff of the department, who helped directly or indirectly for their cooperation and encouragement.

[A. Sravani (Y22ACM406)]

[A. Srilakshmi (Y22ACM402)]

[A. Sai (Y22ACM405)]

[K. Venkata Krishna (Y22ACM447)]

[T. Nagendra (L23ACM501)]

ABSTRACT

Modern software development is increasingly complex, demanding rapid development, high-quality code, and effective collaboration across all phases of the Software Development Life Cycle (SDLC). Code Buddy is an AI-driven system designed to automate software development by transforming natural language instructions into fully functional applications. The system adopts a multi-agent architecture using LangGraph, mimicking a real-world development team. It consists of three core agents: the Planner, which interprets user requirements and generates a structured project plan; the Architect, which breaks the plan into modular tasks and defines the project structure; and the Coder, which generates and organizes the source code.Built on a modern Python environment and integrated with the Groq API, the system enables efficient, scalable, and organized code generation. Experimental results show a significant reduction in development time from several hours to just a few minutes while maintaining consistent performance across various application types. This approach highlights the potential of AI-powered multi-agent systems to streamline planning, design, and coding processes, making software development faster, more efficient, and accessible.

Keywords: Multi-Agent Systems, LangGraph, Groq API, Large Language Models, Automatic Code Generation, Software Development Automation, Natural Language Processing

TABLE OF CONTENTS

ABSTRACT	i
LIST OF TABLES	iv
LIST OF FIGURES	v
CHAPTER 1 – INTRODUCTION	1
1.1 Background of the study	1
1.2 Problem statement	1
1.3 Objectives of the project	2
1.4 Scope of the Project	3
1.5 Organisation of the project	3
CHAPTER 2 – LITERATURE REVIEW	5
2.1 Overview of existing work	5
2.2 Comparisions of previous approaches	5
CHAPTER 3 – SYSTEM ANALYSIS	7
3.1 Existing system	7
3.2 Proposed system	8
3.3 Feasibility study	8
3.4 Requirements	9
CHAPTER 4 – SYSTEM DESIGN	12
4.1 Architecture design	12
4.2 UML diagrams (Use case,Class,Sequence,etc.)	13
4.3 Database Design (ER diagrams,schema)	16
4.4 Algorithm/design methodology	17
CHAPTER 5 – IMPLEMENTATION	18
5.1 Technologies used	18
5.2 Modules description	18
5.3 Code snippets	19
CHAPTER 6 – TESTING	78
6.1 Test cases	78
6.2 Testing methods (Unit,integration,System testing)	78
6.3 Results and analysis	79
CHAPTER 7 – RESULTS AND DISCUSSION	80
7.1 Output screenshots	80
7.2 Performance Analysis	82
7.4 Comparision With Existing System	83
CHAPTER 8 – CONCLUSION AND FUTURE WORK	85
8.1 Summary Of The Project	85
8.2 Achievements	85

8.3 Limitations 85
8.4 Future Enhancements 86
REFERENCES 88

LIST OF TABLES

Table 2.5 Comparative Analysis of Existing AI Development Approaches 8

Table 3.1 Summary of Existing System Limitations and How Code Buddy Addresses Them 12

Table 5.1 Hardware Requirements for the Code Buddy System 17

Table 5.2 Software Requirements 18

Table 7.1 Manual vs AI Development Time Comparison 26

Table 7.2 AI System Success Rate Across Application Types 27

Table 7.3 Effort Distribution in AI-Based Development 28

LIST OF FIGURES

Figure 4.2 Multi-Agent Code Buddy System Architecture 13
Figure 6.1 High-Level System Architecture Diagram 20
Figure 6.3 Data Flow Diagram – Level 0 22
Figure 6.4.1 UML Use Case Diagram 23
Figure 6.4.2 Sequence Diagram – Agent Interaction Flow 24
Figure 6.4.3 UML Class Diagram – Code Buddy Module Structure 25
Figure 7.1 Manual vs AI Development Time Comparison Graph 26
Figure 7.2 AI System Success Rate Across Different Application Types 27
Figure 7.3 Effort Distribution in AI-Based Development (Pie Chart) 28